1 Fig. 1.1 shows sea water flowing down a channel into a tank without splashing. The water is flowing at a rate of 800 kg / min. The length and width of the tank are 3.10 m and 1.20 m. The density of the sea water is 1020 kg / m^3.

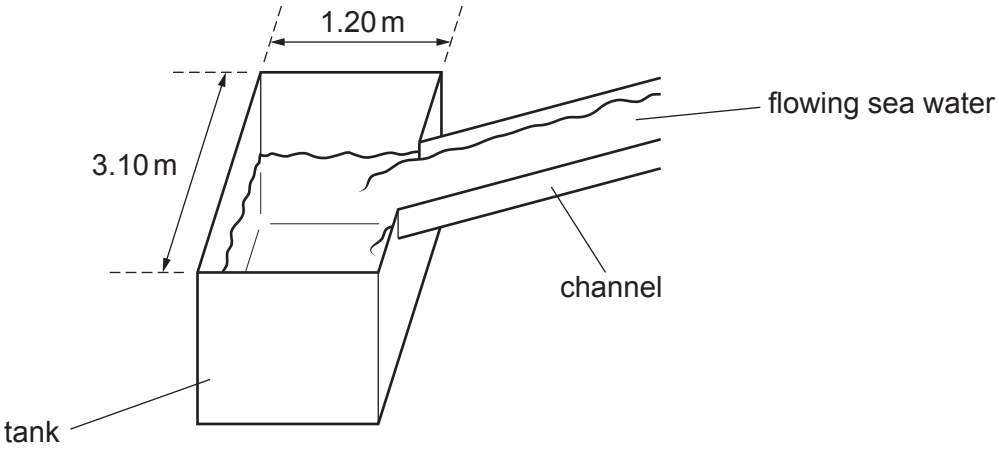


Fig. 1.1 (not to scale)

(a) Initially, the tank is empty.

Calculate the depth of water in the tank after 1.00 minute. Give your answer to three significant figures.

depth = ... [3]

(b) The height of the water decreases by 0.420 m as it flows down the channel.

Calculate the decrease in gravitational potential energy of the water each second.

decrease in gravitational potential energy = ... [3]

(c) The water stops flowing. The depth of water in the tank is 0.800 m.

Calculate the pressure at the bottom of the tank due to the water.

pressure = ... [3]

[Total: 9]

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